Evidence-Grounded Hybrid Retrieval for Document Intelligence

Jane Smith, Rahul Verma. Journal of Applied Information Retrieval. Keywords: hybrid retrieval, knowledge graph, reranking.

Abstract

We propose a hybrid retrieval architecture that combines BM25 lexical search, dense embeddings and knowledge graph expansion with a feature-based reranker. Experiments on a corpus of 1,200 enterprise documents show that Recall@10 improves from 0.71 to 0.86 over a dense-only baseline [1].

1. Introduction

Retrieval-augmented systems often fail because the retrieved evidence is incomplete or untraceable [2]. We present a method in which every claim must cite evidence.

2. Related Work

Prior work on dense passage retrieval [3] and BM25 baselines [4] motivates fusion approaches. Graph-based expansion was studied by Lee et al. [5].

3. Methodology

Our pipeline performs semantic search, lexical search and graph retrieval, merges the candidates and reranks them using seven features. We evaluate with Recall@K, Precision@K and MRR on a held-out dataset. The ablation study removes each retrieval component in turn.

4. Results

The full system achieves MRR of 0.78. Removing graph retrieval reduces Recall@10 by 4 percentage points. Figure 1 summarises the ablation.

5. Conclusion

Hybrid retrieval with reranking improves evidence quality. We release our evaluation dataset.

References

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[2] Lewis et al. Retrieval-Augmented Generation. 2020.

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[5] Lee et al. Graph Expansion for Retrieval. 2022.